

Introduction

Automatic pollination methods like robotic bees utilize vibration to transfer pollen into the pistil of the flower to be pollinated (Figure 1). However, they are recorded to have a lower yield fruit rate compared to the conventional farmed fruits with natural pollination, due to risk of damaging the flower petals.

Spray pollination is established as an effective natural pollination, however is inconsistent due variable wind. Comparing wind-pollinated fruits to vibration pollinated plants, the airflow pollinated plants produce 11.2% fruit yield rate, and the fruits are 20.56% heavier than the vibration pollinated fruits.

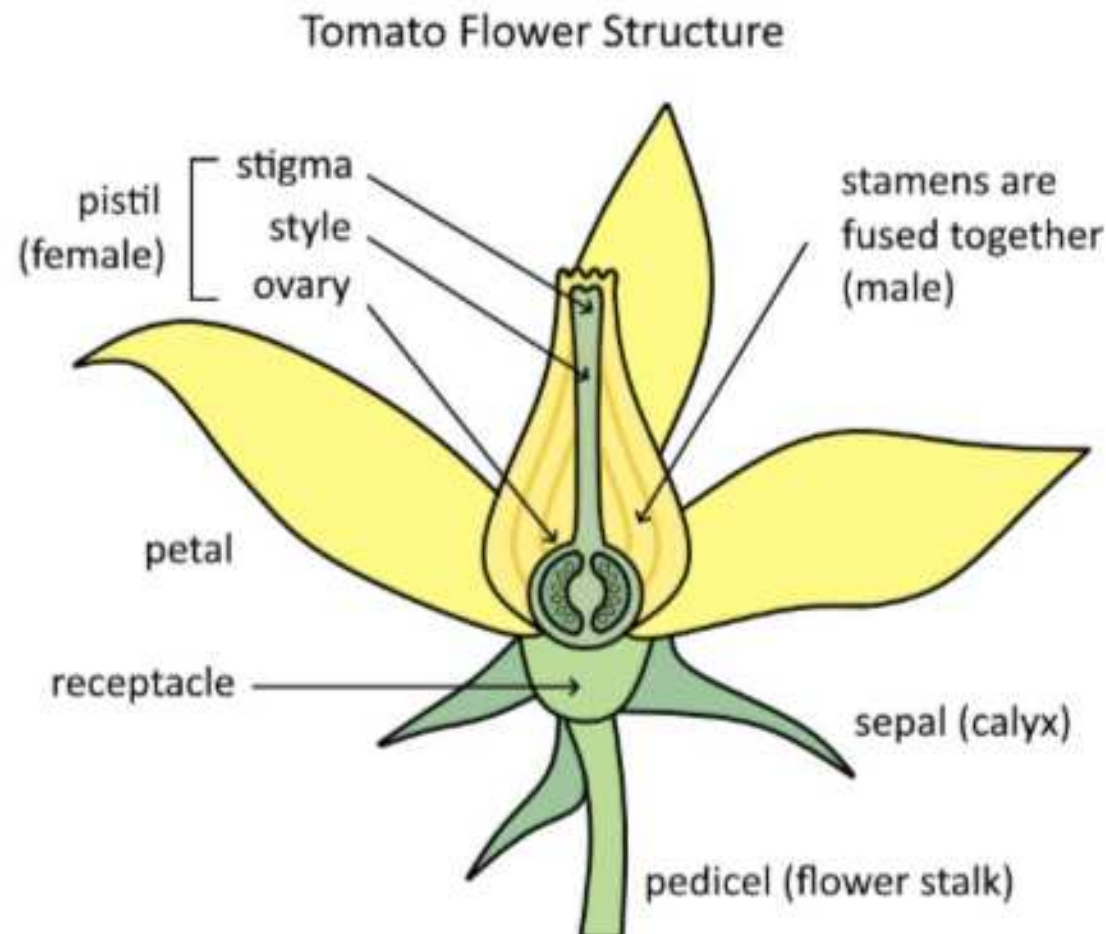


Figure 1

Purpose

Demand for food production is expected to increase by 50% by 2030 (USDA), along with strides in the world population. Food production is significantly strained in resources such as land, and water, as well as expenses in pollination. Natural means of pollination such as insect pollination, which makes up 75% of all pollination is threatened as insect populations have decreased by 45% in the last 40 years (UC Riverside). This significantly spikes the cost of insect pollination, as colonies cost on average \$209 per colonies, which virtually is ineffective to large scale farming operations.

Therefore, vertical farming is being invested in, which significantly cuts the amount of land and resources needed comp However, natural pollination methods like insect pollination are not viable, due to the insects being unable to be established in greenhouse environments. Most greenhouses use manual labor by workers, making operations labor intensive and costly.

Automatic Spray Pollination in Vertical Farming Systems

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1. Built a prototype (Figure 2) that mechanizes the wind pollination, making it able to be implemented in vertical farming environments. The mechanism should make the suck-and-release motion of a syringe bulb connected to a servo-motor operated claw.

2. Wire the servo-motor to an Arduinio board and Breadboard, with the power source being a 9V battery. Program on Arduinio (Figure 3) a simple rotation of the servo-motor to transfer the rotational energy to the claw, causing it to squeeze and release the bulb.

```
10
11
12
13 void loop() {
14   servo1.write(0);
15   delay(3000);
16   servo1.write(90);
17   delay(3000);
18   servo1.write(180);
19   delay(3000);
20 }
```

Figure 3



Figure 2

3. Minimized the total motion of squeezing and releasing the bulb to 15 seconds, allowing intervals to grip and secure the bulb. Optimal angle is found to be 40 degrees with a 60% PWM.

4. Continued prototyping by using PM10 (flour) as a pollen substitute, mechanizing the code to suck and spray the PM10 effectively without leaks.

5. Used PM2.5 (sand) as pollen substitute to be sucked and sprayed unto synthetic flowers, showing the projection and accuracy of the prototype in transferring and ejecting PM2.5 2-5 mms within the pistil area.

Microscopic View of Trials

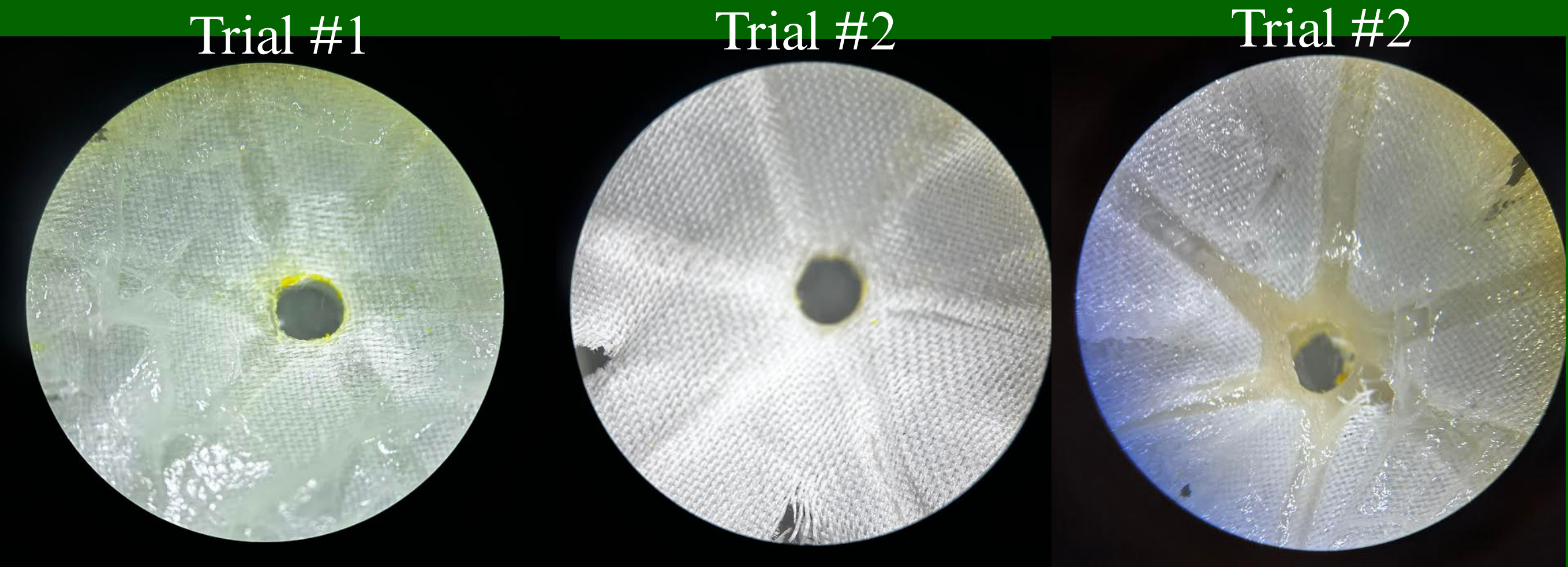


Figure 4

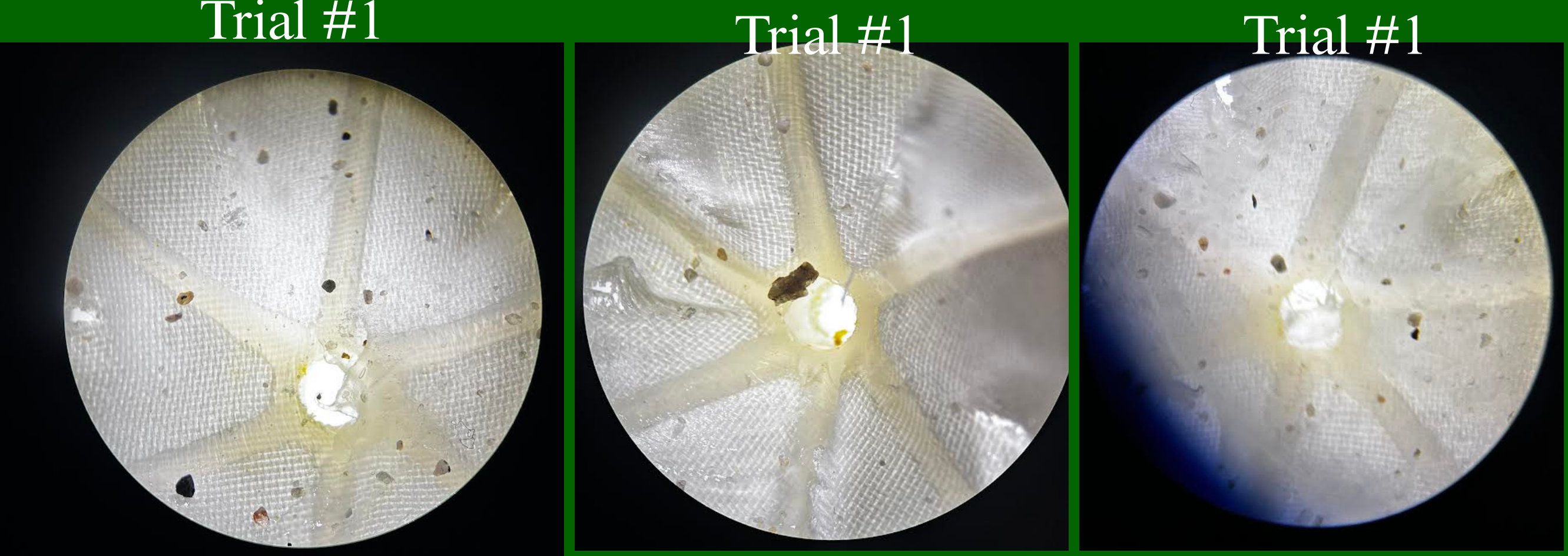


Figure 5

Flower, Trial #	Number of PM2.5 observed
Flower #1	23
Flower #2	11
Flower #3	27

Results

After 3 trials of progressing the prototype with programming, two flower contenders would have been successfully pollinated, with at least 20 particulates ejected unto the pistil area. One flower contender would have eggs fertilized, however not been fully pollinated.

Conclusion

Through the trials done in this study, the prototype is able to pollinate a plant able to a 66.67% success rate. However, due to the small sample size and limited trials, more testing and trials are needed to be done in order to positively quantify its effectiveness. Additionally, if this prototype is to be implemented for pollination for vertical farms, it should be implemented in a overhead system to be able have a line mechanization.

Applications

The device can be implemented in indoor agricultural facilities, greenhouses, and vertical farms to increase fruit yield, quality, and consistency.

By providing a controlled and repeatable method of pollination, the proposed device reduces the need for manual labor and eliminates reliance on unstable insect-based pollination, which has been shown to be difficult to sustain in enclosed environments.

Sources:

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